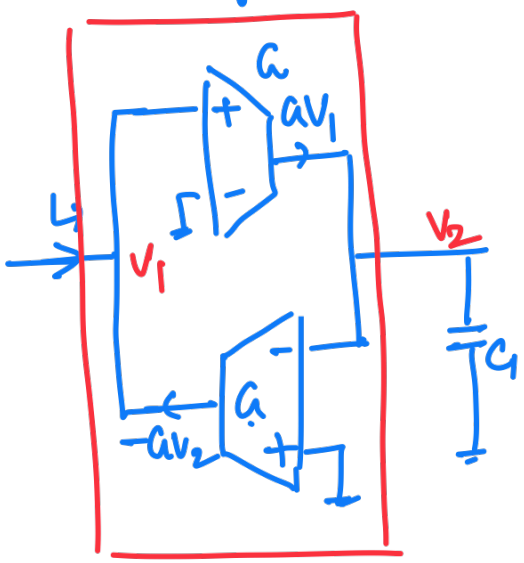
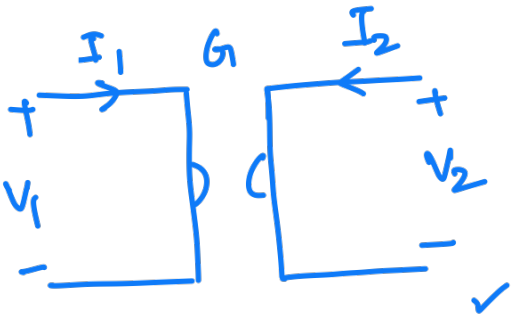
Lecture #31

- Active-RC filter
- Gm-C filter

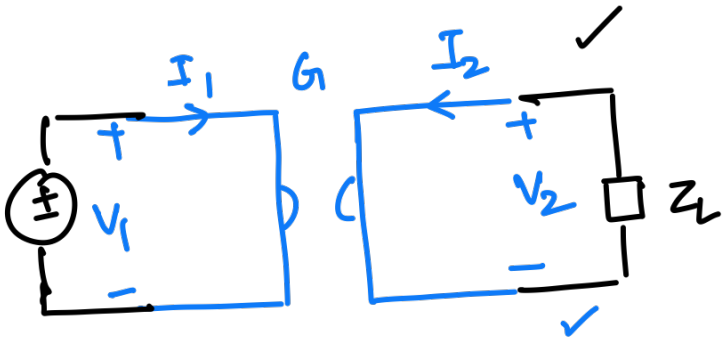


Gyrator (Impedance Inverter)



$I_1 = GV_2$

$I_2 = -GV_1$

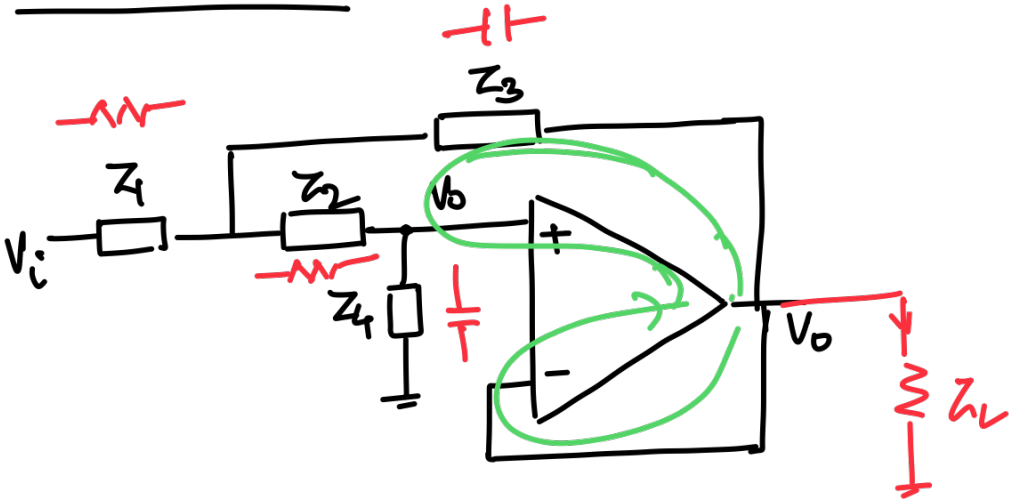


$V_2 = -I_2 Z_L$

$I_2 = -GV_1$

$$\frac{V_1}{I_1} = \frac{V_1}{GV_2} = \frac{V_1}{G(-I_2 Z_L)} = \frac{V_1}{G(-(-GV_1)Z_L)} = \frac{V_1}{(G^2 Z_L)V_1} = \frac{1}{G^2 Z_L}$$

Sallen-Key filter



$$\frac{V_o}{V_i} = \frac{Z_3 Z_4}{Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_3 + Z_3 Z_4}$$

$$= \frac{Y_1 Y_2}{Y_3 Y_4 + Y_2 Y_4 + Y_1 Y_4 + Y_1 Y_2}$$ V✓

$$\left.\frac{V_o}{V_i}\right|_{LPF} = \frac{1}{1 + \frac{s}{\omega_p Q_p} + \frac{s^2}{\omega_p^2}} = \frac{1}{\frac{\omega_p^2 Q_p + \omega_p s + s^2 Q_p}{\omega_p^2 Q_p}}$$

$$= \frac{\omega_p^2 Q_p}{s^2 Q_p + s\omega_p + \omega_p^2 Q_p}$$

$Y_1 = G_1$, $Y_2 = G_2$

$Y_3 = sC_3$, $Y_4 = sC_4$

$$\left.\frac{V_o}{V_i}\right|_{LPF} = \frac{Y_1 Y_2}{Y_3 Y_4 + Y_2 Y_4 + Y_1 Y_4 + Y_1 Y_2}$$

$$= \frac{C_1 C_2}{sC_3 \cdot sC_4 + C_2 \cdot sC_4 + G_1 \cdot sC_4 + C_1 C_2}$$

$$= \frac{1}{\frac{s^2 C_3 C_4}{C_1 C_2} + s\frac{C_4 (C_1 + C_2)}{C_1 C_2} + 1}$$

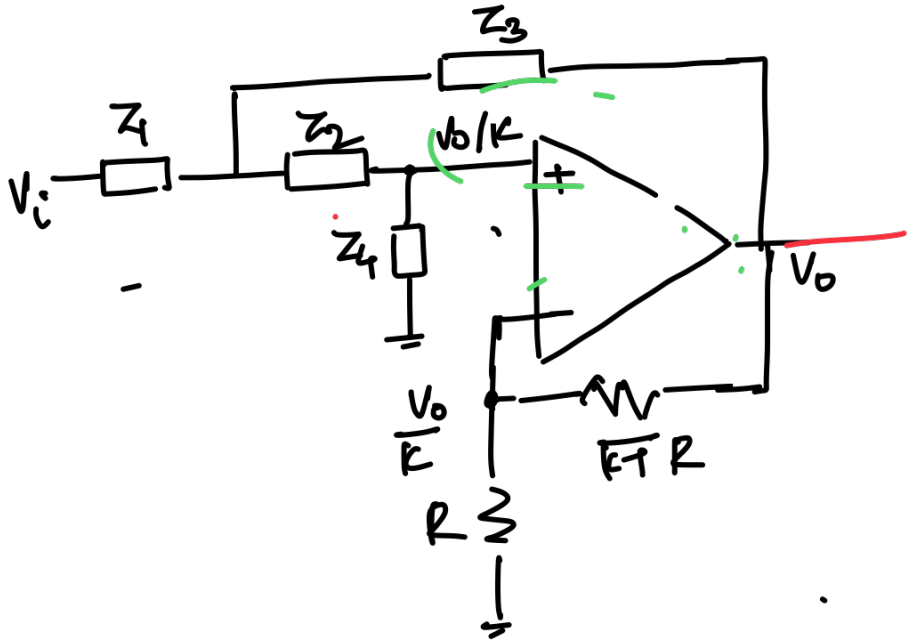
$$\frac{V_o}{V_i} = \frac{Z_3 Z_4}{Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_3 + Z_3 Z_4}$$

$$= \frac{}{\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1}$$

$Z_3 = R_3$, $Z_4 = R_4$, $Z_1 = sL_1$, $Z_2 = sL_2$

$$\frac{V_o}{V_i} = \frac{R_3 R_4}{s^2 L_1 L_2 + sL_1 R_3 + sL_2 \cdot R_3 + R_3 R_4}$$

$$-\left.\frac{V_o}{V_i}\right|_{s=0} = ?$$



$$\frac{V_o}{V_i} = \frac{K Z_3 Z_4}{Z_3 Z_4 + Z_1 Z_4 (1-K) + Z_1 Z_3 + Z_1 Z_2 + Z_2 Z_3}$$

$Z_1 = R_1$, $Z_2 = R_1$ $Z_3 = \frac{1}{sC_3}$, $Z_4 = \frac{1}{sC_4}$

$$= \frac{K \frac{1}{s^2 C_3 C_4}}{\frac{1}{s^2 C_3 C_4} + \frac{R_1}{sC_4}(1-K) + \frac{R_1}{sC_3} + R_1 R_2 + \frac{R_2}{sC_3}}$$

$$= \frac{(K)}{1 + sC_3 R_1 (1-K) + sC_4 R_1 + s^2 C_3 C_4 R_1 R_2 + sC_4 R_2}$$

$$= \frac{(K)}{s^2 C_3 C_4 R_1 R_2 + s\left(\underbrace{C_4 R_2 + C_4 R_1 + R_1 C_3 (1-K)}_{-m}\right) + 1}$$

$$\left.\frac{V_o}{V_i}\right|_{s=0} = K$$

$$s_1 + s_2 = \frac{-b}{a}$$

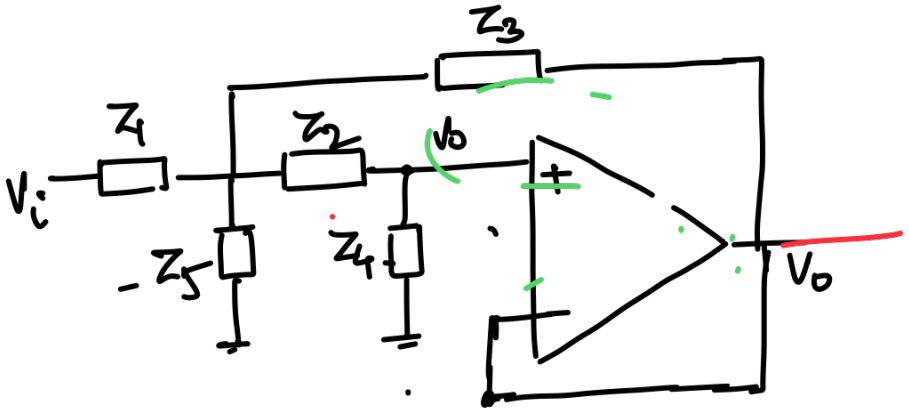
$$C_4 R_2 + C_4 R_1 + R_1 C_3 (1-K) \geq 0$$

$$\frac{R_2}{R_1} + 1 + \frac{C_3}{C_4}(1-K) \geq 0$$

$$\frac{C_4}{C_3} \cdot \frac{R_2}{R_1} + \frac{C_4}{C_3} + 1 \geq K$$

$$K \leq \frac{C_4}{C_3} \cdot \frac{R_2}{R_1} + \frac{C_4}{C_3} + 1$$

Bandpass Salen-Key filter.



$$\frac{V_o}{V_i} = \frac{Z_3 Z_4 Z_5}{Z_1 Z_3 Z_4 + Z_1 Z_3 Z_5 + Z_1 Z_2 Z_3 + Z_1 Z_2 Z_5 + Z_3 Z_4 Z_5 + Z_2 Z_3 Z_5}$$ ✓

$$= \frac{Y_1 Y_2}{Y_2 Y_5 + Y_2 Y_4 + Y_4 Y_5 + Y_3 Y_4 + Y_1 Y_2 + Y_1 Y_4}$$ ✓

$$\frac{V_o}{V_i} = \frac{s C_2 G_1}{s^2 C_2 C_5 + s C_2 G_4 + s C_5 G_4 + G_3 G_4 + s C_2 G_1 + G_1 G_4}$$

$Y_1 = G_1$, $Y_2 = sC_2$, $Y_3 = G_3$, $Y_4 = G_4$, $Y_5 = sC_5$

✓ ✓ ✓

$$= \frac{s C_2 G_4}{s^2 C_2 C_5 + s(C_2 G_4 + C_5 G_4 + C_3 G_4 + C_2 G_1) + G_1 G_4}$$

"Bi-Quad"

$$\frac{\alpha}{\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1}$$

Rauch-filter

